

PSL Recognition — SignVLM Signer-Disjoint Finetune Report

Run: `signVLM_lists` artifacts | 46 epochs | Best checkpoint: epoch 45 (val loss 0.1967, val acc 94.61%*)

**In-training validation pass (per-epoch table, §2). The independent post-training re-evaluation of this same checkpoint (Final metrics, §2) measures 94.53% — see protocol note below.*

All numbers, curves, and confusion matrices in this report are reconstructed directly from the logged artifacts in `docs/retrain_details/signVLM_lists/` (`signVLM_loss_history.json`, `signVLM_step_log.csv`, `signVLM_final_metrics.csv`, `signVLM_full_metrics_test.csv`, `*_confusion_matrix.npy`). Every figure is regenerable with `generate_report_assets.py`; this PDF is built by `build_pdf.py`.

1. Model architecture

SignVLM — frozen CLIP image backbone + lightweight EVL-style temporal decoder + linear classifier.

Stage	Layer	Notes
backbone	CLIP ViT-L/14 (Inpre variant), frozen	per-frame visual features, 224×224 input
temporal decoder	4 transformer decoder layers, qkv_dim 1024, 16 attention heads	EVL-style, attends across frames
head	Linear → 104 logits	PSL-104 classifier

Input: 16–24 frames/clip, 224×224, RGB. **Classes:** 104 (PSL-104 — English gloss words + Urdu alphabet letters; `label_map_auto.json`).

For reference, the base SignVLM architecture (Luqman, 2025, *PeerJ Comput. Sci.* 11:e3112) reports **58.6 M trainable parameters in the EVL-style temporal decoder + classification head** — the CLIP backbone is frozen and contributes 0 trainable parameters, so it is not part of that count. No parameter-count snapshot was captured for this specific PSL-104 run.

Training recipe

Setting	Value	Source
Train / Val / Test split	4,368 / 1,244 / 1,248 clips	<code>train.tsv</code> / <code>val.tsv</code> / <code>test.tsv</code> (<code>unseen.tsv</code> empty, not evaluated)
Test protocol	signer-disjoint (held-out signer), n=1,248	<code>signVLM_full_metrics_test.csv</code> (<code>test_diff_signer</code>)
Batch size	20 (inferred: 218 steps/epoch × 20 ≈ 4,368 with <code>drop_last</code>)	<code>signVLM_step_log.csv</code>
Optimizer	AdamW	training log
Learning rate	4e-5 initial, cosine-annealed to ~1e-8	<code>signVLM_loss_history.json</code>
Scheduler	CosineAnnealingLR (T_max = num_steps)	training log
Loss	CrossEntropyLoss	training log
Weight decay	0.05 (project default, not confirmed in this run's logs)	project scripts
Epochs completed	46 (<code>global_step</code> 0–10,027, 218 steps/epoch)	<code>signVLM_step_log.csv</code>
Checkpointing	best-val-loss checkpoint saved at the every-5-epoch evals (epochs 5, 10, ..., 45)	training log

Key protocol notes

- **Train accuracy is train-mode / augmented** (`Acc1(aug)`), so it *understates* the model's true fit — final eval-mode train accuracy is 98.01% vs. the ~85% seen in the curves.
- **Validation ran every 5th epoch** (epochs 5, 10, ..., 45) — the val curves have points only at those epochs.
- **Final metrics are eval-mode, multi-view** for train/val (`signv1m_final_metrics.csv`).
- **The epoch-45 val acc in the per-epoch table (94.61%) and the Final-metrics val acc (94.53%) are both multi-view eval on the same best checkpoint, run as two independent passes** — one logged live inside the training loop, one re-run afterward by a separate script. Multi-view eval samples frames with a random component not pinned to a fixed seed across runs, so the two passes disagree by exactly 1 of 1,244 clips (1,177 vs. 1,176 correct). This is evaluation-pass noise, not a train/eval-mode mismatch or a copy-paste error, and it is well within run-to-run variance for a sample this size.
- The test split is the **same 104 classes performed by a signer never seen in training** (signer-disjoint protocol).

2. Per-epoch metrics

epoch	lr	train_loss	train_acc1	train_acc5	val_loss	val_acc1	val_acc5
1	4.00e-05	4.9275	0.87%	4.76%			
2	3.97e-05	4.8302	1.15%	5.41%			
3	3.94e-05	4.7591	1.26%	6.06%			
4	3.90e-05	4.3192	6.90%	19.50%			
5	3.85e-05	3.1776	26.42%	52.87%	1.5456	64.31%	91.16%
6	3.80e-05	2.5564	39.70%	65.60%			
7	3.73e-05	2.2036	49.98%	70.60%			
8	3.66e-05	1.4651	65.48%	82.68%			
9	3.58e-05	1.2812	70.76%	84.38%			
10	3.49e-05	1.2110	71.74%	85.39%	0.4107	89.95%	99.20%
11	3.39e-05	1.1317	73.23%	86.24%			
12	3.29e-05	1.0185	75.41%	87.98%			
13	3.18e-05	1.0011	76.01%	88.12%			
14	3.06e-05	0.9231	78.21%	88.76%			
15	2.94e-05	0.9422	77.34%	88.44%	0.2844	92.77%	99.36%
16	2.82e-05	0.8700	79.04%	89.70%			
17	2.69e-05	0.8645	78.97%	88.92%			
18	2.56e-05	0.8386	79.43%	89.77%			
19	2.42e-05	0.8014	80.07%	89.79%			
20	2.28e-05	0.8046	80.48%	89.98%	0.2502	92.44%	99.52%
21	2.14e-05	0.7697	80.69%	91.06%			
22	2.01e-05	0.7225	81.74%	91.65%			
23	1.87e-05	0.7445	81.28%	91.24%			
24	1.73e-05	0.6844	82.66%	91.86%			
25	1.59e-05	0.6901	82.82%	91.70%	0.2245	94.05%	99.20%
26	1.45e-05	0.6532	83.17%	92.82%			
27	1.32e-05	0.6913	82.34%	91.83%			
28	1.19e-05	0.6498	83.07%	92.61%			
29	1.07e-05	0.6205	84.29%	92.55%			
30	9.46e-06	0.6273	83.90%	92.82%	0.2070	94.21%	99.36%
31	8.30e-06	0.6208	83.99%	92.39%			
32	7.20e-06	0.5953	84.47%	93.37%			
33	6.16e-06	0.5884	84.84%	93.28%			
34	5.19e-06	0.6166	83.81%	92.66%			
35	4.29e-06	0.5771	85.00%	93.58%	0.2049	94.53%	99.20%
36	3.47e-06	0.6047	84.68%	93.14%			
37	2.72e-06	0.5744	85.30%	92.96%			
38	2.06e-06	0.5980	84.66%	92.84%			
39	1.49e-06	0.5288	86.15%	94.17%			

40	1.01e-06	0.5956	84.31%	93.17%	0.1974	94.61%	99.44%
41	6.18e-07	0.5586	85.30%	93.85%			
42	3.22e-07	0.5738	85.50%	93.85%			
43	1.21e-07	0.5682	85.25%	93.49%			
44	1.62e-08	0.5718	85.14%	93.78%			
45	8.65e-09	0.5907	84.72%	93.39%	0.1967	94.61%	99.36%
46	8.20e-09	0.5853	85.07%	93.56%			

Val columns are blank on epochs where validation was not scheduled; accuracy values are top-1/top-5 percentages of the split. Train accuracy is train-mode on augmented clips.

Final metrics (eval mode, multi-view) — best model

Split	n	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	Precision (wtd)	Recall (wtd)	F1 (wtd)
train	4,368	98.01%	0.9811	0.9801	0.9800	0.9811	0.9801	0.9800
validation	1,244	94.53%	0.9504	0.9454	0.9457	0.9503	0.9453	0.9456

Test (signer-disjoint, held-out signer, 1,248 clips) — best model

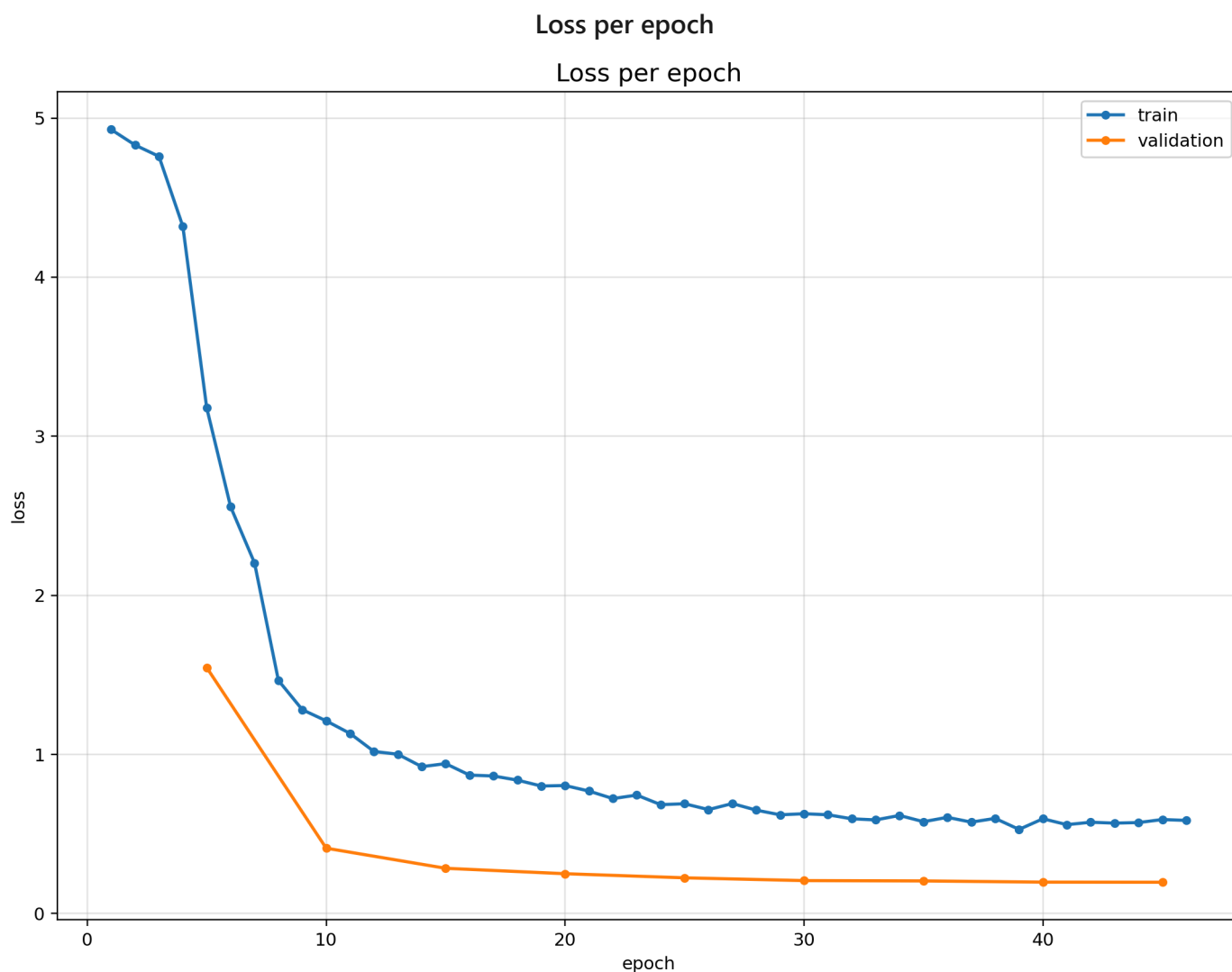
Metric	Value
test_top1_acc	78.12%
test_precision (macro)	0.8150
test_recall (macro)	0.7813
test_f1 (macro)	0.7713
test_precision (weighted)	0.8150
test_recall (weighted)	0.7813
test_f1 (weighted)	0.7713

Exact value: $975/1,248 = 78.125\%$, rounded to 78.12% throughout this report — a few earlier drafts rounded this figure up to 78.13%; that has been corrected for consistency with the charts and confusion-matrix titles in §3–5.

Macro and weighted precision/recall/F1 are identical to four decimal places on the test split because the test set is exactly balanced — 1,248 clips over 104 classes = 12 clips/class — so the per-class weighting used by "weighted" averages collapses to the unweighted mean used by "macro" averages. This is expected for a balanced split, not a duplicated column.

A second held-out test set drawn from the **same signer pool** as train/val (`test_confusion_matrix.npy` , n=1,248) scores **93.75%** top-1 — so the pure signer-shift cost is 93.75% → 78.12% (–15.6 points), not 94.5% → 78.1%.

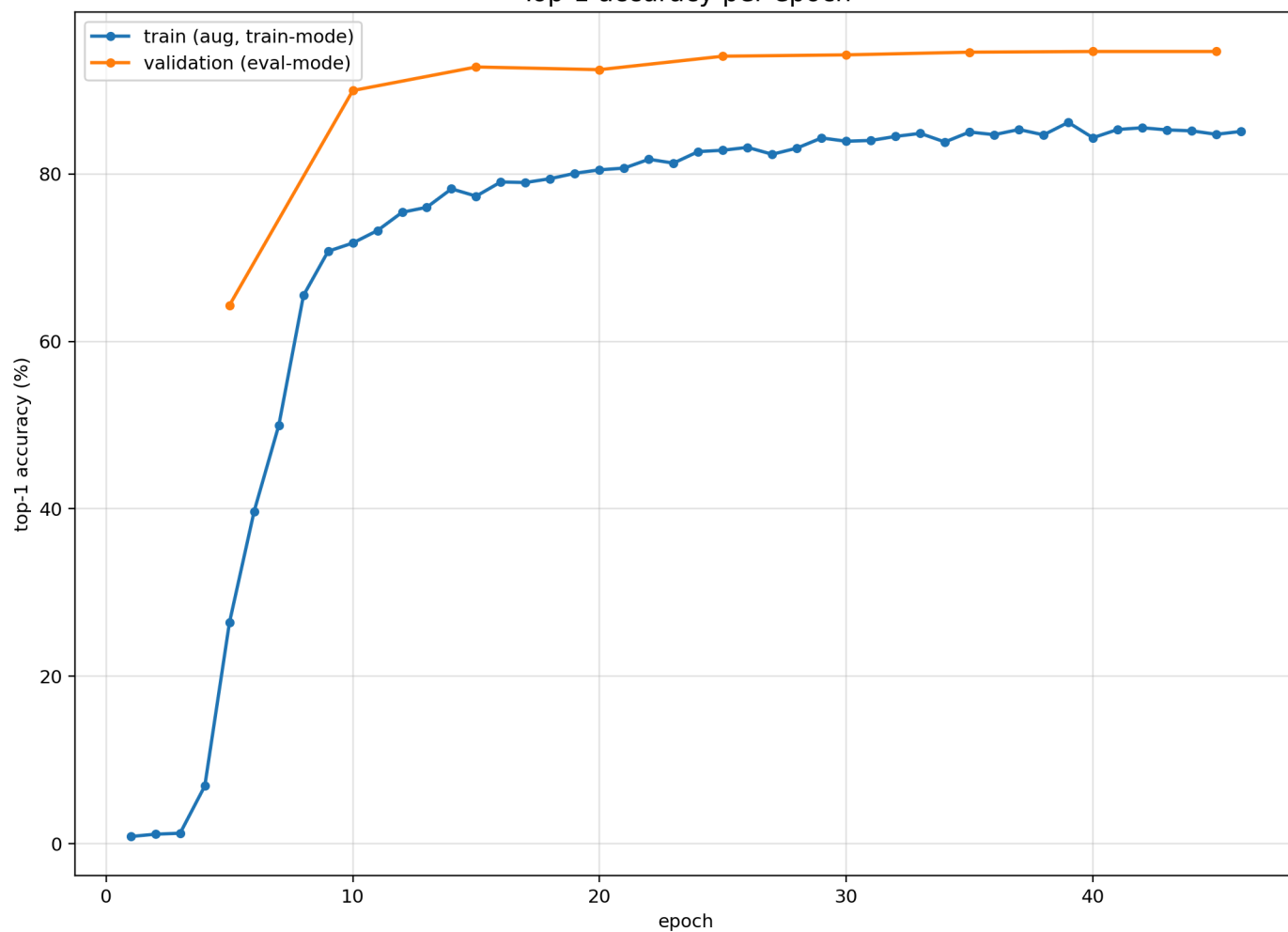
3. Training curves



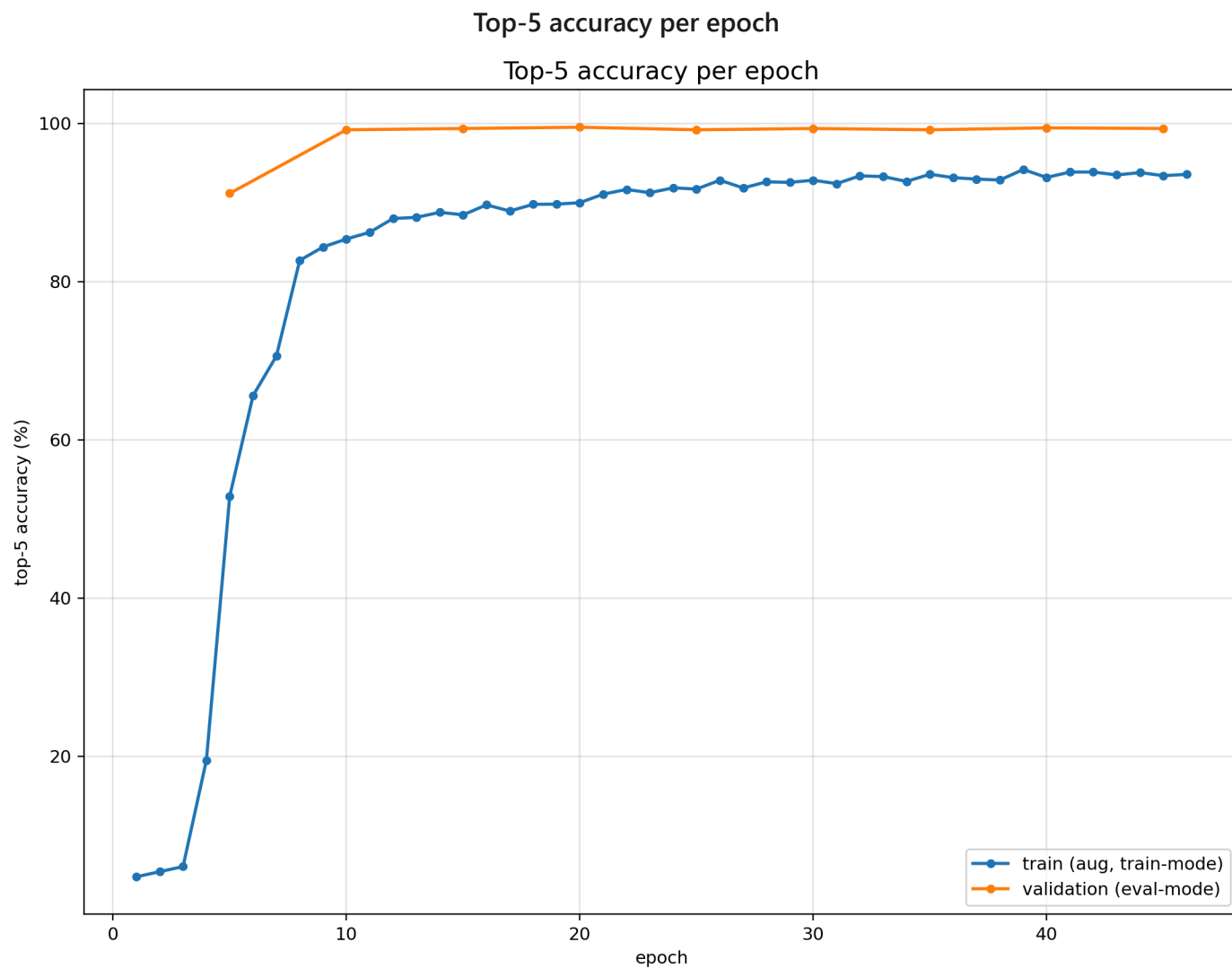
Cross-entropy loss for train (blue) and validation (orange) over all 46 epochs; validation is evaluated every 5th epoch. Train loss falls from 4.93 to 0.59. Validation loss decreases monotonically at every scheduled evaluation (1.55 at epoch 5 $\rightarrow$ 0.1967 at epoch 45) and never inflects upward — there is no overfitting signature anywhere in the run. Validation sits below train because train loss is measured in train mode on augmented clips with dropout active.

Top-1 accuracy per epoch

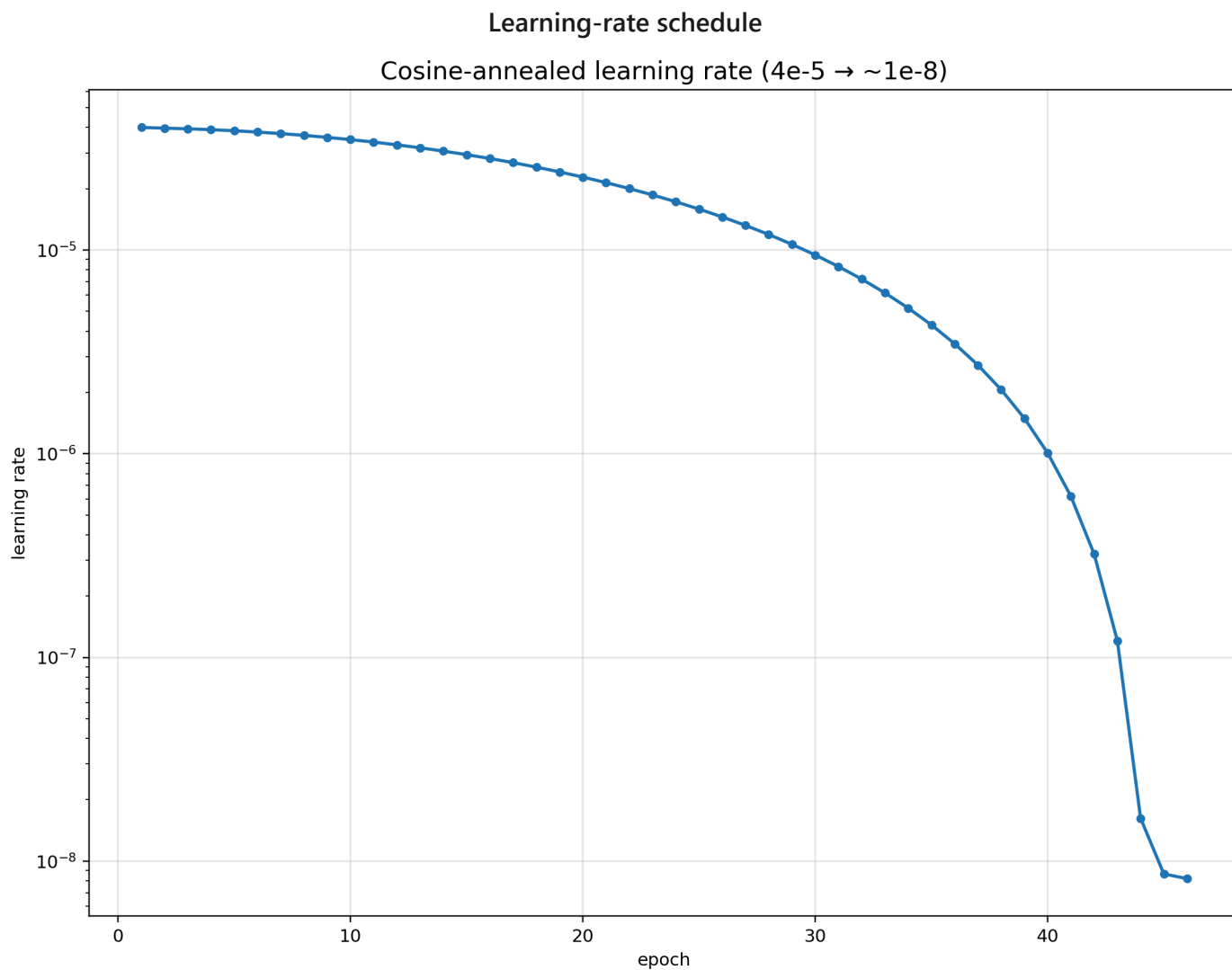
Top-1 accuracy per epoch



Top-1 accuracy for train (blue, train-mode on augmented clips) and validation (orange, eval-mode). Validation reaches 89.95% by epoch 10 and saturates near 94.6% from epoch 35 onward. The validation curve sitting above train is expected under this protocol: the train series is depressed by augmentation and active dropout, while eval-mode train accuracy on the final model is actually 98.01%.



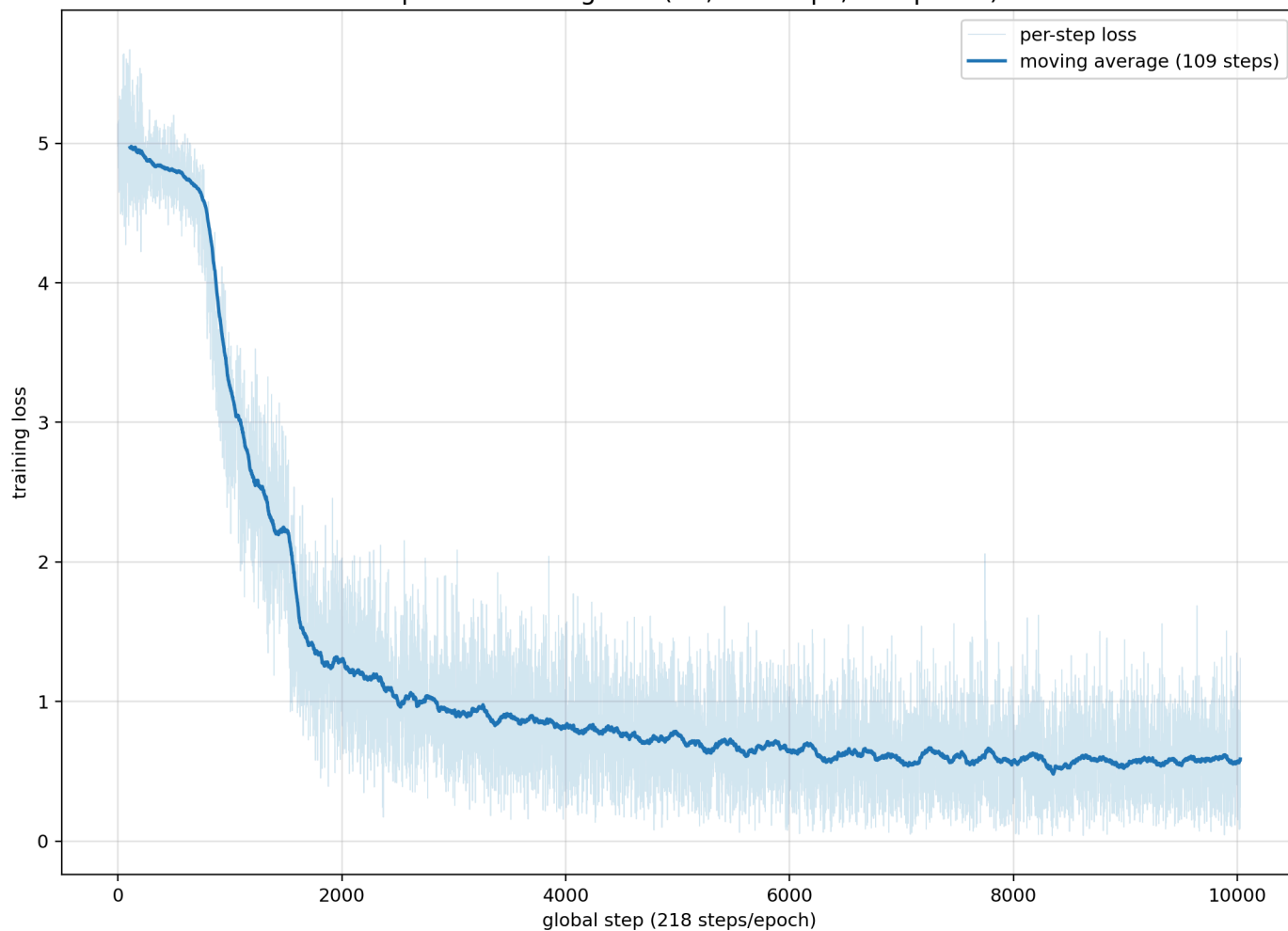
Top-5 accuracy for train and validation. Validation top-5 exceeds 99% from epoch 10 onward and ends at 99.36% — for nearly every validation clip the correct sign is within the model's five most confident predictions, meaning residual top-1 errors are mostly fine-grained confusions between similar signs rather than outright misses.



Cosine-annealed learning rate over the run (log scale): $4\text{e-}5$ at epoch 1 decaying smoothly to $\sim 1\text{e-}8$ by epoch 46, with no warmup phase and no restarts. The smooth decay matches the plateauing of the loss and accuracy curves over the final third of training.

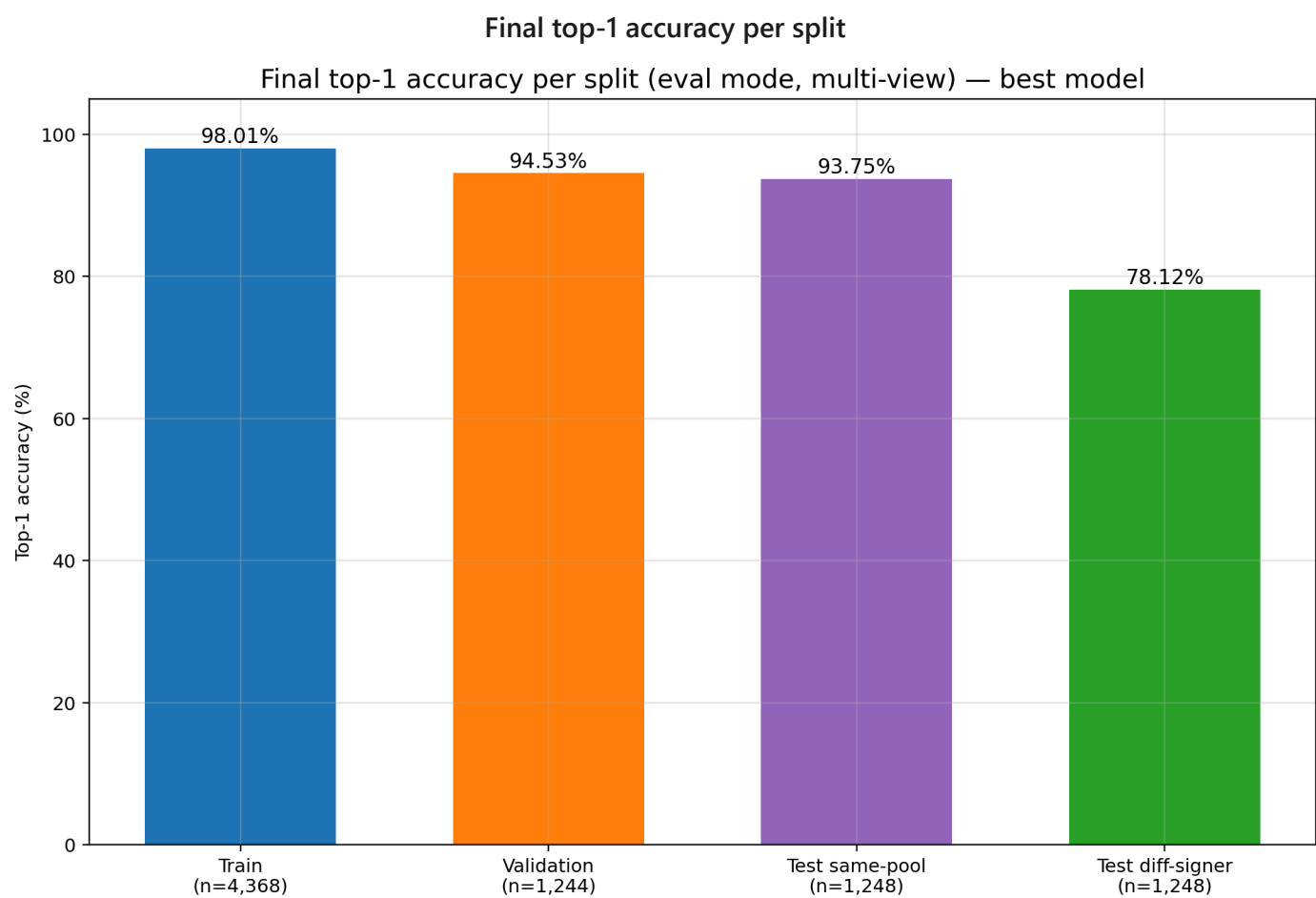
Step-level training loss

Step-level training loss (10,028 steps, 46 epochs)



Per-step training loss across all 10,028 optimizer steps (light blue, batch size 20, 218 steps/epoch) with a 109-step (~half-epoch) moving average (dark blue). The sharp drop between steps ~800 and ~2,000 (epochs 4–9) is where the temporal decoder locks onto the frozen CLIP features; afterwards the average declines gradually to ~0.6 with well-behaved per-batch variance and no instability spikes.

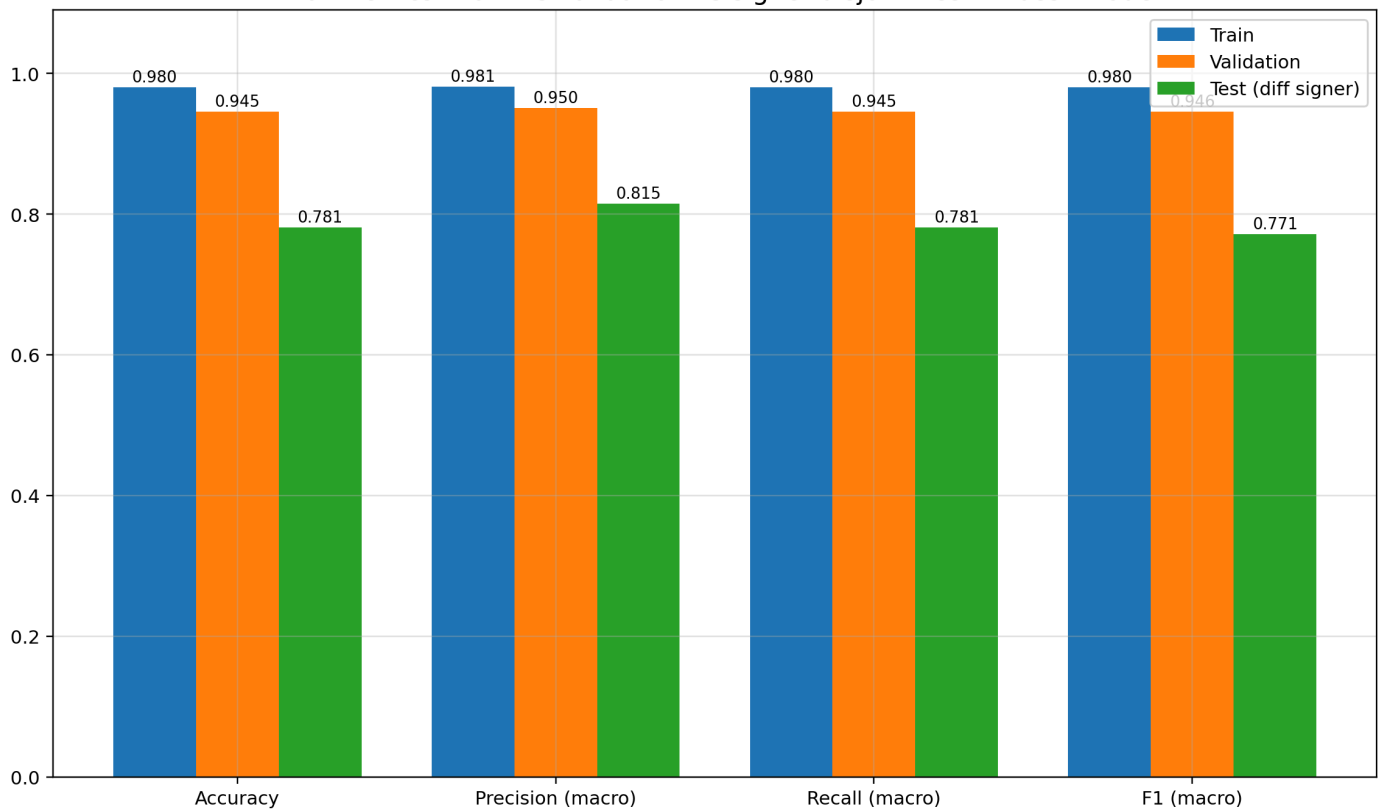
4. Split-level results



Final-model top-1 accuracy (eval mode, multi-view) on all four splits. Train 98.01% and validation 94.53% come from signvml_final_metrics.csv; the two test bars are computed from their confusion matrices. The key comparison is the last two bars: a held-out test set from the same signer pool scores 93.75%, while the signer-disjoint test (a person never seen in training) scores 78.12% — isolating a –15.6-point cost attributable purely to signer shift.

Final metrics — train vs validation vs signer-disjoint test

Final metrics: train vs validation vs signer-disjoint test — best model

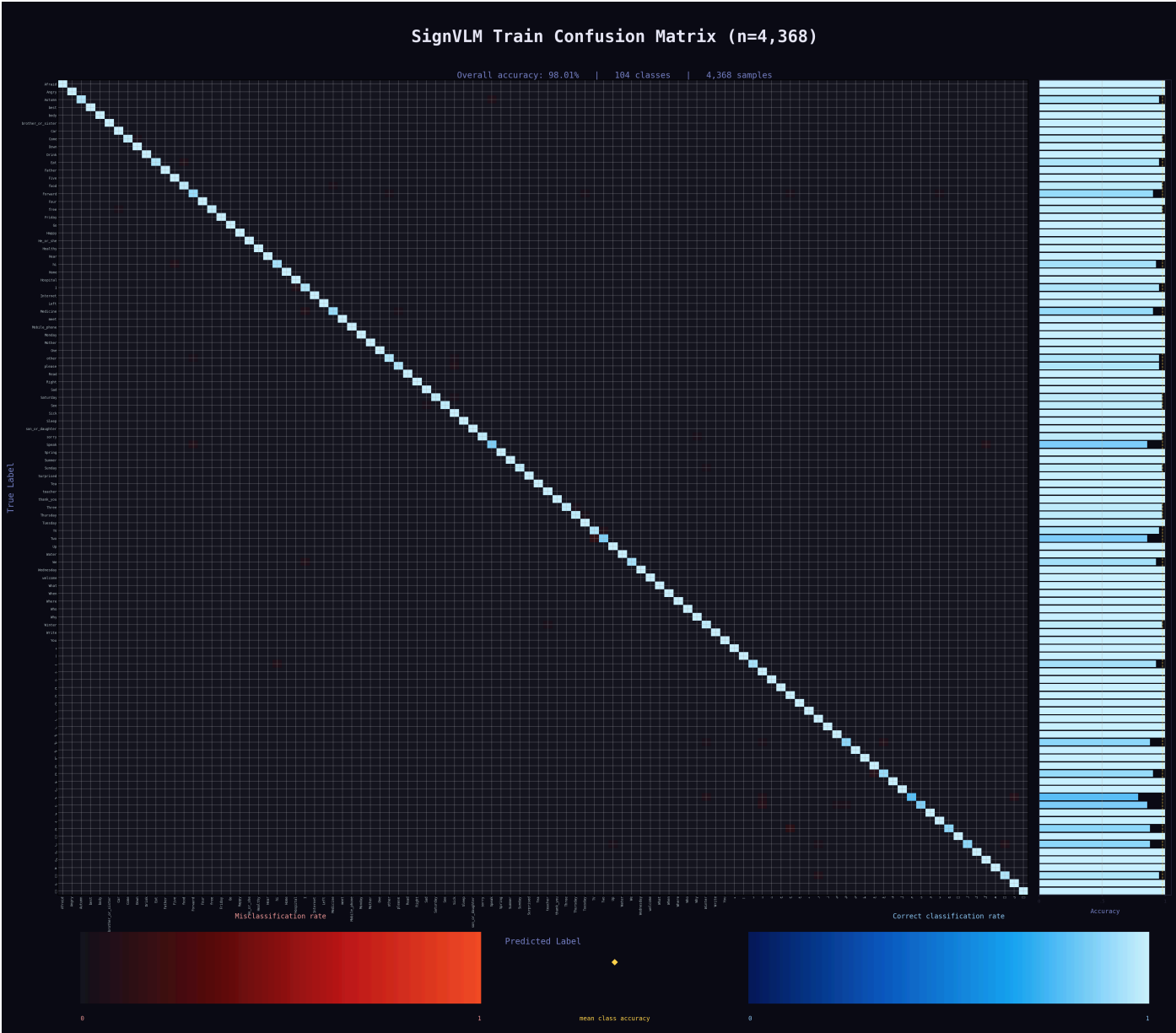


Accuracy, macro precision, macro recall, and macro F1 for train, validation, and the signer-disjoint test (best model). The three metrics degrade together and moderately (accuracy $0.980 \rightarrow 0.945 \rightarrow 0.781$; F1 $0.980 \rightarrow 0.946 \rightarrow 0.771$), and macro precision (0.815) exceeding accuracy on the test split indicates errors are concentrated in a minority of classes rather than spread uniformly.

5. Confusion matrices

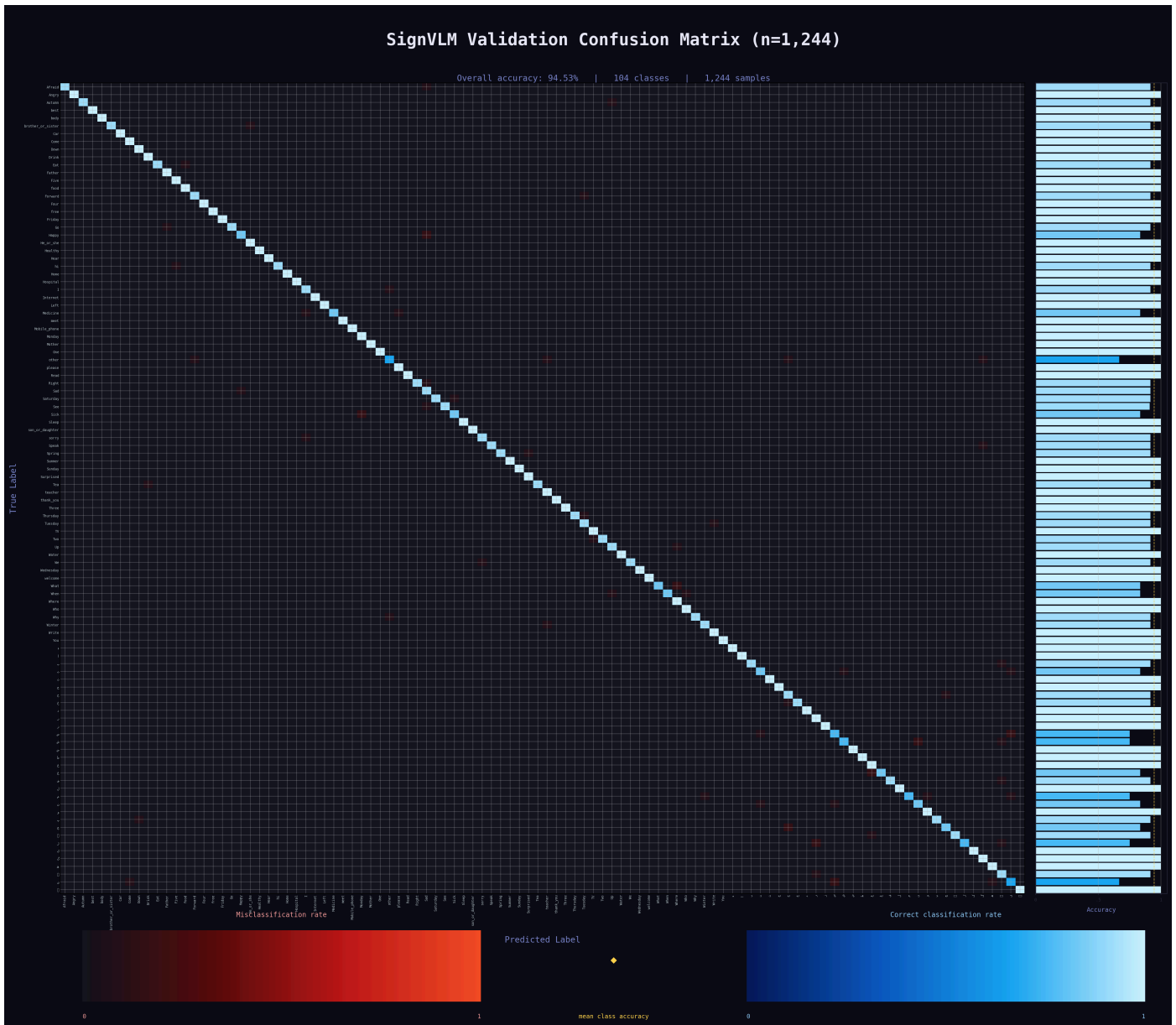
Rendered with the project's `confusion_matrices/confusion_matrix_builder.py`. How to read it: rows are true classes, columns are predicted classes, row-normalized. The blue diagonal encodes each class's correct-classification rate (brighter = higher); red off-diagonal cells are misclassifications (brighter red = larger error mass). The right-hand strip shows per-class accuracy as horizontal bars, with the yellow dashed line marking mean class accuracy.

Train confusion matrix (n=4,368, acc 98.01%)



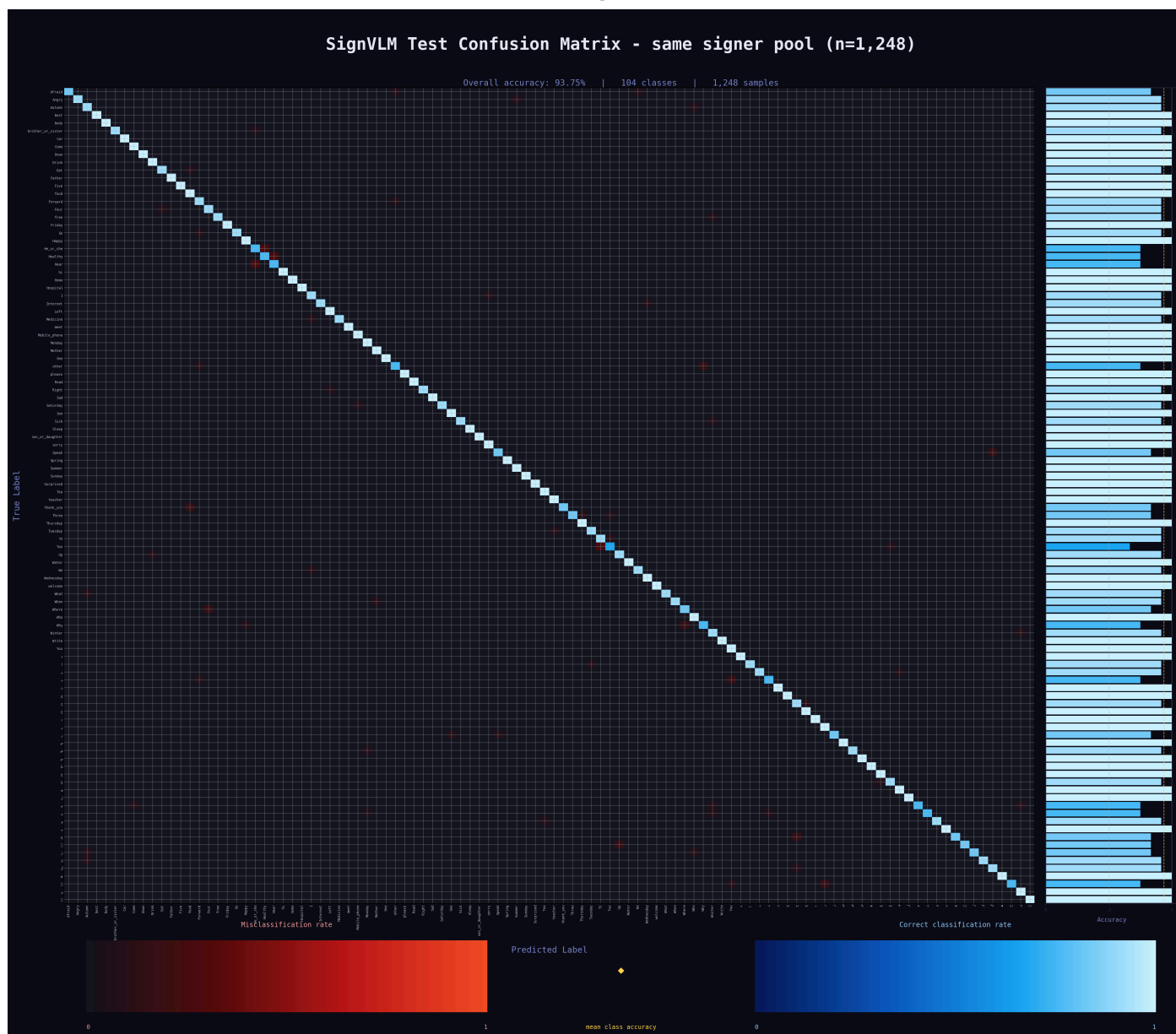
Final-model predictions on the training split (eval mode). The diagonal is nearly uniformly bright — 4,281 of 4,368 clips correct — with only faint scattered off-diagonal cells, confirming the model has essentially fit the training pool without residual class-level blind spots.

Validation confusion matrix (n=1,244, acc 94.53%)



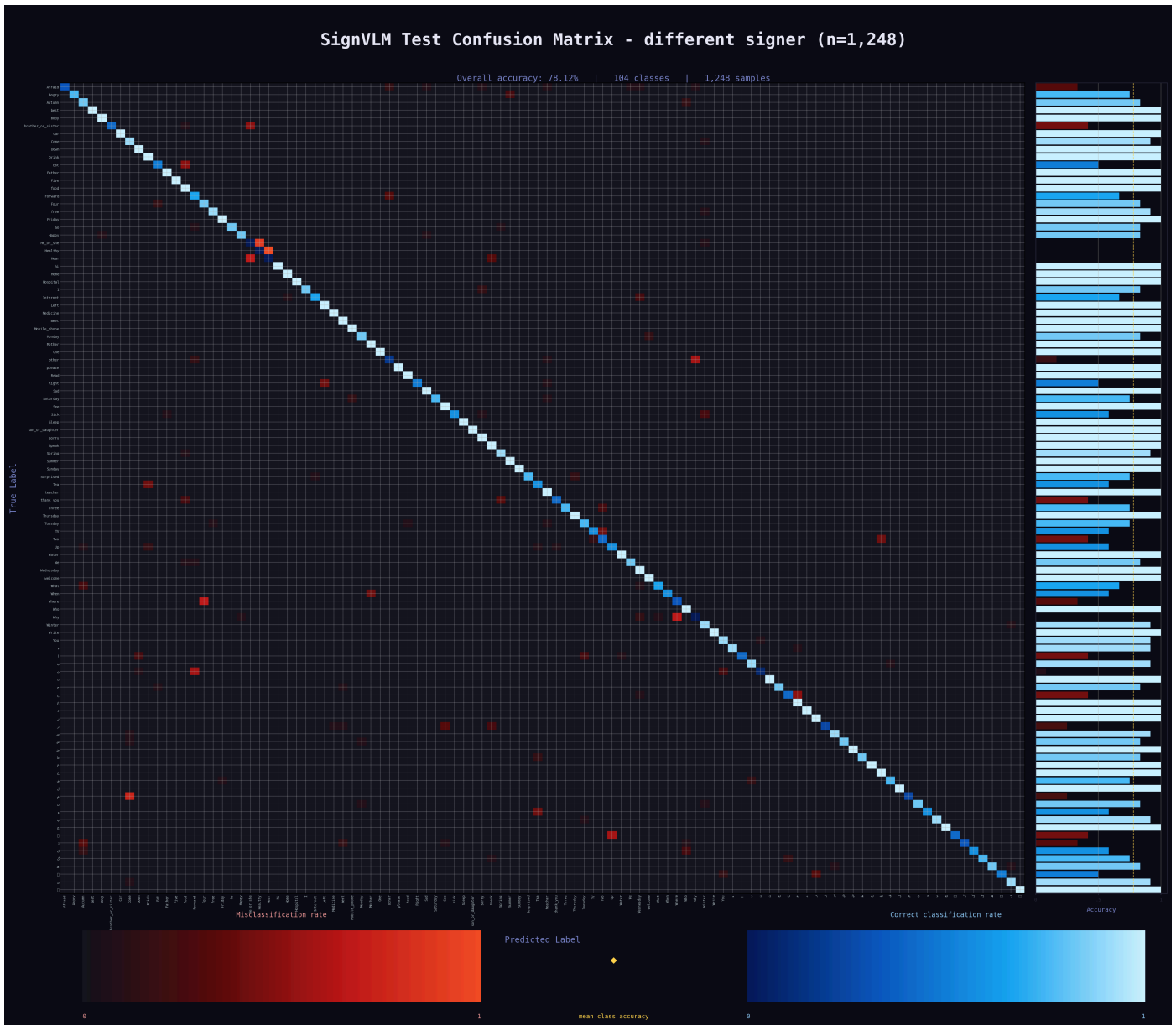
Predictions on the validation split (same signer pool as train). The structure mirrors the train matrix with slightly more off-diagonal mass (1,176/1,244 correct). Per-class accuracy bars on the right stay high across nearly all 104 classes — there is no class the model systematically fails within the seen-signer distribution.

Test confusion matrix — same signer pool (n=1,248, acc 93.75%)



Held-out test clips drawn from the same signers as train/val. Performance is statistically indistinguishable from validation (93.75% vs 94.53%), which establishes the baseline the signer-disjoint matrix below should be compared against: any additional degradation there is attributable to the new signer, not to held-out-clip noise.

Test confusion matrix — different signer (n=1,248, acc 78.12%)



The core result: predictions on 1,248 clips from a signer never seen during training. 45 of 104 classes remain perfect, and 55 (including those 45) score $\geq 90\%$ overall, but errors concentrate in structured cliques of visually similar signs — Hear/Healthy/He_or_she form a near-closed confusion cycle (hand-near-face signs), Why $\rightarrow$ Where and Where $\rightarrow$ Four confuse adjacent question/counting signs, and $\rho \rightarrow$ Come is a systematic Urdu-letter/gloss collision. This structured failure pattern is the signature of genuine sign similarity under a new signer's motion style.

Where the diff-signer errors concentrate

Per-class accuracy on the signer-disjoint test: **45/104 classes are perfect (100%)**, and **55/104 classes score $\geq 90\%$ overall** (this 55 includes the 45 perfect classes — i.e. 10 more classes land in the 90–99% band), mean class accuracy 78.12% — but 17 classes fall below 50%, and four collapse completely. (The remaining $104 - 55 - 17 = 32$ classes fall in the 50–89% band.)

Class	Acc	Correct/Total	Confused with (count)
Hear	0%	0/12	He_or_she (8), Speak (4)
He_or_she	0%	0/12	Healthy (11), Winter (1)
Healthy	0%	0/12	Hear (12)

Why	0%	0/12	Where (8), Wednesday (2)
ت	8%	1/12	Forward (7), You (3)
other	17%	2/12	Why (7), Forward (2)
ج	25%	3/12	See (4), Speak (3)
م	25%	3/12	Come (9)
Afraid	33%	4/12	other (2), Why (1)
Where	33%	4/12	Four (8)

6. Read

- **Training is healthy end-to-end.** Train loss 4.93 → 0.59; eval-mode train accuracy reaches 98.01%. Val loss decreases monotonically at every scheduled eval (1.55 at epoch 5 → 0.1967) and never inflects upward — no overfitting signature over 46 epochs.
- **Most of the fit happens by epoch 10** (val 89.95% top-1, 99.20% top-5); the remaining 36 epochs of cosine decay add ~4.7 points of val accuracy and steadily better calibration (val loss 0.41 → 0.20).
- **The signer-disjoint generalization gap is moderate.** SignVLM goes 94.53% val → 78.12% test on a signer never seen in training. Against the same-pool test set (93.75%) the pure signer-shift cost is -15.6 points — a real, measurable gap, but the model retains the large majority of its skill on a person it has never seen.
- **Residual errors are concentrated and interpretable** (Section 5): four classes absorb a third of the total error mass via mutual confusion among visually adjacent signs (Hear ↔ Healthy ↔ He_or_she; Why → Where → Four). Top-5 accuracy on val is ~99.4%, and the diff-signer confusions are predominantly within these small cliques.
- **What could push past 78%:**
 1. **More signers in train** — the gap that remains is still signer-identity, and the confusion cliques are exactly where a second/third signer's motion variance would help most.
 2. **Targeted disambiguation of the confusion cliques** — hand-crop or higher-frame-rate sampling for the hand-near-face cluster (Hear/Healthy/He_or_she), which differs mainly in fine hand shape and contact point.
 3. **Light backbone adaptation** (last-block or LoRA finetuning of CLIP) once more signer diversity exists — with a single training-signer pool, unfreezing now would mostly re-open the door to signer memorization.

Report and figures generated 2026-07-19 from the artifacts in `docs/retrain_details/signVLM_Lists/`.